BondBreak Strategy Router

Local pipeline runner for Modules -1–3 with clear, auditable routing output.

Pipeline Output

PIPELINE DECISION

LOW_CONF_GO

PASS | Mode standard | M2 PASS | M3 PASS | TopK 5

Job Type	Selected Bond	Primary Route
STANDARD_TRANSFORMATION	1-3	serine_hydrolase

Difficulty	Compute Active	Scaffolds
EASY	true	60

SCIENTIFIC REPORT

PIPELINE HEALTH & INTERCONNECTION

MODULE	STATUS	RESULT	ATTENTION
Module -1	PASS	ester	OK
Module 0	LOW_CONF_GO	serine_hydrolase	Route confidence low
Module 1	PASS	Top 5 scaffolds	Retention weak
Module 2	PASS	scaffold_lib_serine_hydrolase_v1_scaffold_015	OK
Module 3	PASS	Plan 0.291 / Phys 0.409	OK

Working now

- Module -1 reactivity: PASS (constraint: ELECTROSTATIC)
- Routing accepted (LOW_CONF_GO)
- Bond resolved: 1-3
- Module 1 passed (5 scaffolds ranked)
- Module 2 selected scaffold_lib_serine_hydrolase_v1_scaffold_015
- Negative control assigned
- Physics prior P=0.41

Results snapshot

- Module -1: PASS (constraint: ELECTROSTATIC)
- Route: serine_hydrolase
- Module 1: PASS
- Module 2: scaffold_lib_serine_hydrolase_v1_scaffold_015
- Module 3 arms: 5
- Neg control: A4

Needs improvement

- Route confidence low
- Retention score low

Why this score

- Route confidence 0.41 below 0.60 (Route confidence below this indicates weak mechanistic support.)
- Target resolution 0.93 above 0.85 (Target resolution below this indicates ambiguous bond selection.)
- Retention 0.34 below 0.45 (Retention below this implies weak binding/orientation risk.)
- Physics prior P=0.34, k_eff=2.14e-03 s^-1
- Plan phys 0.41, SNR=22.5

Handoff checks

- OK M-1 → M0: resolved_target
- OK M0 → M1: selected_bond
- OK M1 → M2: top_scaffolds
- OK M2 → M3: module3_handoff
- OK Shared IO: condition_profile + telemetry

CONDITION FLOW (SHARED IO)

pH	Temp C	Ionic strength
7.25	30	n/a

Solvent	Cofactors	Run ID
n/a	n/a	35dfdf37-285b-4f8c-ad79-6e4c134f6bc4

MODULE	GIVEN EFFECT	OPTIMUM	CONFIDENCE	RETRY
Module 0	route P=0.80	pH 6.5-8, T 25-37C	0.80 (Low)	Close to optimum
Module 1	topology 0.45	pH 6.50, T 25.0C	0.45	Close to optimum
Module 2	k_pred 3.6168	pH 6.50, T 37.0C	0.61	pH 6.88, T 33.5C
Module 3	Baseline activity under user conditions.	n/a	0.31	not_applicable

SHARED IO DETAILS

Trace	module0, module1, module2, module3
Bond Spec	acetyl_ester_C-O

EVIDENCE LOG (CLOSED LOOP)

Load Download

Per-module datapoints recorded for this run.

Run ID: 35dfdf37-285b-4f8c-ad79-6e4c134f6bc4

No evidence loaded yet.

CONFIDENCE

Route confidence 0.41 (42%)

Target resolution 0.93 (93%)

Wetlab (target) 0.63 (63%)

Overall signal 0.24 (24%)

Wet-lab priors are shown for context, not gating.

INTENT & ROUTING

Requested Output
salicylic acid

Intent Type
hydrolysis

Intent Conf
0.80

Requires Trap
false

Mechanism route

- Primary: serine_hydrolase
- Secondary: metallo_esterase
- Mechanisms: acyl-oxygen hydrolysis

Expert tracks

- serine_hydrolase (weight 0.9, required false)
- metallo_esterase (weight 0.4, required false)

Success definition

- Any activity: Detectable conversion or substrate turnover under plausible conditions.
- Target specific: Product-specific conversion consistent with requested transformation under plausible conditions; not optimized for industrial yield.

MECHANISM CONTRACT

Route ID
serine_hydrolase

Reaction family
hydrolysis

Expected nucleophile
Ser

Allowed geometries
serine_og

Policy
KEEP_WITH_PENALTY

Mechanism match

- Status: UNVERIFIED
- Expected: Ser
- Observed: n/a
- Penalty: n/a

UNITY ARBITRATION

Stage
module3

Recommendation
monitor

Penalty
0.100

Hard veto
false

Weights

- physics: 0.406
- chem: 0.130
- bio: 0.290
- math: 0.174

MODULE 0 PHYSICS AUDIT

Temp K
303.15

DeltaG_dagger (kJ/mol)
88.04

Eyring k (s^-1)
0.0043

Horizon (s)
3600

Prior success P
0.339

Physics-based prior when historical routing data is sparse.

TARGET RESOLUTION

Selection Mode
token

Canonical Token
ester__acyl_o__acetyl

Match Count
1

Candidate Mode
token

Scanned Bonds
12

Candidate Matches
1

Selected bond

Bond	1-3
Role	ester__acyl_o
Role Conf	0.90

BOND	ROLE	SCORE	STATUS
1-3	ester__acyl_o	1.30	Selected

MODULE -1 SRE

Status	Bond Type	Primary Role
PASS	ester	ester

Constraint	Confidence	Cache
ELECTROSTATIC	0.70	Miss

Attack sites

Electrophile	1
Leaving	3
Stabilizable	2

Eligibility

Serine hydrolase	APPROVED

Route bias

Prefer	serine_hydrolase, metallo_esterase
Discourage	None
Strength	0.20

SRE CPT SIGNALS

CPT	DELTA	DIRECTION	METHOD
None			

Delta uses relative CPT scale; see raw JSON for detail.

STRUCTURE CONTEXT

Atoms	Heavy Atoms	Rings
13	13	1
Hetero Atoms	**Bond Class**	**Bond Type**
4	ester	ester

Size proxies

Approx radius	1.20
Approx volume	7.24
Min diameter	2.40

Bond center hint

Atoms	1, 3
Anchor	C[C:1](=O)[O:2]

Functional groups

- carboxylic_acid: 1
- aryl_ring: 1
- ester: 1
- acid: 1

COMPUTE & CONSTRAINTS

Scaffolds	TopoGate	QM
60	standard	none
MD	**Review**	**Active**
none	true	true
pH Min	**pH Max**	**Temp C**
6.5	8	30
Metals	**Oxidation**	**Host**
true	true	n/a

Assumptions

- metals_allowed default true
- oxidation_allowed default true
- constraints defaulted from route

MODULE 1 SUMMARY

Status
PASS

Mode
standard

Weights
access 0.35, reach 0.45, retention 0.20

Cache Hits
37

Cache Misses
23

Cache Size
939

SCAFFOLD	TOTAL	ACCESS	REACH	RETENTION
scaffold_lib_serine_hydrolase_v1_scaffold_015	0.479	0.498	0.528	0.335
scaffold_lib_serine_hydrolase_v1_scaffold_043	0.399	0.498	0.346	0.344
scaffold_lib_serine_hydrolase_v1_scaffold_047	0.391	0.674	0.202	0.323
scaffold_lib_serine_hydrolase_v1_scaffold_034	0.388	0.595	0.438	0.259
scaffold_lib_serine_hydrolase_v1_scaffold_053	0.327	0.652	0.230	0.265

Showing top 5 scaffolds.

Physics audit

Temp K
303.15

DeltaE model (kJ/mol)
9.10

Boltz mean
0.016

Boltz min/max
0.007–0.037

Diffusion cap
1000000000.0000

MODULE 2 SUMMARY

Status	Selected Scaffold	Adjusted Score
PASS	scaffold_lib_serine_hydrolase_v1_scaffold_015	2.476

Retention Risk	Nucleophile Geo	Best Variant
MEDIUM	serine_og	Oxyanion hole strengthening

Mechanism Consistency
Primary hypothesis: serine nucleophile Mechanism match: UNVERIFIED

Objective

Improve retention without breaking access or reach.

Selection rationale

Selected scaffold_lib_serine_hydrolase_v1_scaffold_015 with ranking score 2.47587. Predicted k_mean=3.61682 (CI90 0.11058-12.69365) using Eyring ranking. Retention penalty applied (MEDIUM retention risk).

Retention penalty reasons

- MEDIUM retention risk

Physics audit

DeltaG_dagger base (kJ/mol)	DeltaDeltaG_dagger (kJ/mol)	Strain (kJ/mol)
88.04	-2.00	1.50

DeltaG_dagger variant (kJ/mol)	k variant (s^-1)
87.54	0.0052

Module 2 warnings

- W_DECISION_LOW_CONF_GO: proceeding with caution.
- W_MECH_UNVERIFIED: No detected nucleophile from scaffold residues. (confidence penalty 0.10).

Module 2 errors

- None

VARIANT PROPOSALS

VARIANT	SCORE	MUTATIONS	RATIONALE
Oxyanion hole strengthening	1.000	{"target_role":"oxyanion_hole","preferred_residues":["Ser","Thr","Asn","Gln"],"note":"strengthen tetrahedral intermediate stabilization"}	Improve stabilization of the tetrahedral intermediate.
Polar anchor (carboxylate)	0.800	{"target_role":"polar_anchor","preferred_residues":["Arg","Lys","His"],"note":"anchor deprotonated carboxylate"}	Carboxylate is negative at pH 6.5–8; add a cationic anchor.
Baseline	0.800	None	Used as control for downstream comparisons.
Hydrophobic clamp	0.520	{"target_role":"pocket_lining","preferred_residues":["Phe","Tyr","Leu","Ile"],"note":"hydrophobic clamp near aromatic ring"}	Retention weakness; add hydrophobic wall for ring packing.
Access-preserving clamp	0.400	{"target_role":"deep_pocket_lining","preferred_residues":["Tyr","Phe"],"note":"clamp deeper in pocket to preserve access"}	Long tunnel; improve retention without blocking access.
pH tuning bundle	0.400	{"target_role":"general_base","preferred_residues":["His","Asp"],"note":"add His-Asp dyad to tune pKa"}	pH tuning to activate catalytic residues under fixed conditions.

MODULE 3 EXPERIMENT PLAN

Status	Batch ID	Plan Score
PASS	8c33779b-ff09-4141-8f7f-e3254caffd37	0.291

Plan Phys	QC Status	Learning
0.409	N/A	PLANNED

Arms
5

Baseline expected outcome

Baseline activity under user conditions.

Baseline conditions

pH 7.25, T 30.0C

Arms

- A4 • negative_control • no_enzyme • pH 7.25, T 30.0C
- A5 • stress_boundary • Oxyanion hole strengthening • pH 8.00, T 30.0C
- A1 • baseline • Oxyanion hole strengthening • pH 7.25, T 30.0C
- A2 • improved_conditions • Oxyanion hole strengthening • pH 6.88, T 33.5C
- A3 • variant_disambiguation • Oxyanion hole strengthening vs Polar anchor (carboxylate) • pH 7.25, T 30.0C

Information gain notes

- total_eig=0.150814

Physics audit

Horizon (s)	Expected signal	Noise floor
3600	0.450	0.020

SNR
22.50

MODULE 3 QC

QC Status	Records	Router Updates
N/A	5	0

QC reasons

- None

DIAGNOSTICS

Required next input

- None

Protonation flags

- {"group":"carboxylic_acid","count":1,"estimated_pka_range":[3,5],"risk_level":"medium","note":"Carboxylic acids deprotonate near pH 4-5."}

Descriptor status

- None

DATA QUALITY

Warnings

- None

Errors

- None